

Universal dense-open windows for generic conservative surface semigroups and graph-directed local systems

Yitzchak Shmalo

Einstein Institute of Mathematics, The Hebrew University of Jerusalem

Givat Ram, Jerusalem, Israel

yitzchak.shmalo@gmail.com

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Abstract

We prove a universal dense-open window theorem for generic conservative pairs on a closed orientable surface: there exists a residual set of area-preserving pairs (g_0, g_1) such that for *every* choice of dense open sets D_0, D_1 , the restrictions $g_0|_{D_0}$ and $g_1|_{D_1}$ already generate an open-set transitive pseudosemigroup admitting a residual set of dense admissible branches. The residual set of pairs does not depend on the windows.

We then prove a deterministic propagation theorem for graph-directed local systems: once a distinguished core atom carries a transitive pair and every other atom is connected to and from the core through dense open transport windows, dense admissible branches propagate throughout the network. Combining these results with a fibered genericity argument yields a universal parameter theorem for conservative graph-directed systems equipped with two loop words having private single-use edges, removing both the dependence on prescribed windows and any simplicity requirement on the loop words.

1 Introduction

For a continuous self-map on a second-countable Baire space, classical Baire-category arguments relate topological transitivity to the existence of a residual set of points with dense forward orbit; see, for example, [15, 1]. Closely related residual-orbit theorems for countably generated pseudogroups are also classical; compare the work of Epstein–Millet–Tischler and Ghys, as well as the exposition in [2, §8.4]. The first purpose of this paper is to isolate the exact *forward-branch* formulation needed for graph-directed local dynamics and to give a short self-contained proof in that language.

Let (S, ω) be a closed orientable surface and let $\text{Diff}_\omega^r(S)$ denote the space of C^r diffeomorphisms preserving ω . Koropecski and Nassiri proved that for every $r \in \mathbb{N} \cup \{\infty\}$ there exists a residual subset of $\text{Diff}_\omega^r(S) \times \text{Diff}_\omega^r(S)$ consisting of pairs whose iterated-function-system action is transitive on S [13]. The 2011 erratum corrects part of the higher-genus fiberwise discussion in [13]; in the present paper we use only the pair theorem itself as a black-box input [14].

The main new dynamical consequence extracted here is a *universal dense-open window principle*. If (g_0, g_1) is a Koropecski–Nassiri generic conservative pair, then for *every* pair of dense open subsets $D_0, D_1 \subset S$, the restricted partial maps $g_0|_{D_0}$ and $g_1|_{D_1}$ still generate an open-set transitive pseudosemigroup on S , and indeed admit a residual set of starting points with dense admissible branches. Thus generic global conservative transitivity descends automatically to genuinely local window dynamics as soon as the windows are dense open.

This window step is useful for graph-directed systems built from local transition maps. Such systems occur naturally in return-map and scattering-map formalisms in Hamiltonian dynamics [6, 7, 9]. In a different direction, local and graph-directed iterated-function systems have mainly been studied from attractor-theoretic, contractive, or fractal-interpolation viewpoints; see, for instance, Hutchinson, Barnsley, Mauldin–Williams, Massopust, Barnsley–Hegland–Massopust, Barnsley–Vince, and the recent foundational work of Oliveira and Varandas [11, 3, 17, 16, 4, 5, 18]. Our setting is different: the branches are local homeomorphisms (and in the conservative applications, local area-preserving diffeomorphisms), there is no contraction hypothesis, and the goal is topological transitivity and residual dense branching rather than attractor theory.

The main results are:

- a forward-branch extraction of the classical pseudogroup Baire theorem for countable families of partial homeomorphisms (Theorem 2.6);
- a dense-open restriction principle for globally defined semigroup actions (Corollary 3.3);
- a *universal* dense-open window theorem for generic conservative pairs on a closed surface (Theorem 5.3);
- a deterministic accessibility theorem propagating dense branches from a partially defined transitive core through a graph-directed window network (Theorem 4.5);
- residual forcing results for conservative graph-directed systems, including a universal private-edge parameter theorem for two loop words with private single-use edges (Theorem 6.5, Corollary 6.6, and Corollary 6.7).

The paper is organized so that the only genuinely conservative input is the Koropecski–Nassiri theorem. Everything else is a topological argument about partial homeomorphisms, dense-open restrictions, and accessibility in graph-directed local systems.

2 A forward-branch Birkhoff theorem for countable partial homeomorphisms

We begin with the partial-action tool that underlies the whole paper. For background on pseudogroups we refer to [2]; here we only need countable families of partial homeomorphisms and their forward compositions.

Let X be a topological space.

Definition 2.1 (Partial homeomorphism). A *partial homeomorphism* of X is a homeomorphism

$$h : U \rightarrow V$$

between open sets $U, V \subset X$. We write $\text{dom}(h) = U$ and $\text{im}(h) = V$.

Definition 2.2 (Generated forward pseudosemigroup). If $\mathcal{H}$ is a countable family of partial homeomorphisms of X , we write $\langle \mathcal{H} \rangle^+$ for the set of all finite nonempty compositions of elements of $\mathcal{H}$ with nonempty domain.

Definition 2.3 (Open-set transitivity). The pseudosemigroup $\langle \mathcal{H} \rangle^+$ is *open-set transitive* if for every pair of nonempty open sets $A, B \subset X$ there exists $h \in \langle \mathcal{H} \rangle^+$ such that

$$h(A \cap \text{dom}(h)) \cap B \neq \emptyset.$$

Definition 2.4 (Dense admissible branch). An *admissible branch* for $\mathcal{H}$ starting at $x \in X$ is an infinite sequence $h_1, h_2, \dots \in \mathcal{H}$ such that each successive composition is defined at x . We say that x *admits a dense admissible branch* if some admissible branch starting at x has dense orbit in X .

Lemma 2.5 (Residual pullback under a partial homeomorphism). *Let X be a Baire space, let $R \subset X$ be residual, and let $h : U \rightarrow V$ be a partial homeomorphism of X . Then*

$$(X \setminus U) \cup h^{-1}(R)$$

is residual in X .

Proof. Because R is residual, $X \setminus R$ is meager in X . Hence

$$U \setminus h^{-1}(R) = h^{-1}(V \setminus R)$$

is meager in U , and therefore meager in X . But

$$X \setminus ((X \setminus U) \cup h^{-1}(R)) = U \setminus h^{-1}(R),$$

so the displayed set has meager complement and is therefore residual. $\square$

Theorem 2.6 (Forward-branch Birkhoff theorem for countable partial homeomorphisms). *Let X be a second-countable Baire space and let $\mathcal{H}$ be a countable family of partial homeomorphisms of X . If $\langle \mathcal{H} \rangle^+$ is open-set transitive, then there exists a residual subset $R_\infty \subset X$ such that every $x \in R_\infty$ admits a dense admissible branch.*

Proof. Set $\Sigma := \langle \mathcal{H} \rangle^+$. Since $\mathcal{H}$ is countable, so is Σ . Let $\{V_n\}_{n \in \mathbb{N}}$ be a countable basis of X . For each n define

$$Q_n := \bigcup_{h \in \Sigma} \{x \in \text{dom}(h) : h(x) \in V_n\}.$$

Each Q_n is open. It is also dense: if $A \subset X$ is nonempty and open, then by open-set transitivity there exists $h \in \Sigma$ such that $h(A \cap \text{dom}(h)) \cap V_n \neq \emptyset$, so $A \cap Q_n \neq \emptyset$.

Therefore

$$R_0 := \bigcap_{n \in \mathbb{N}} Q_n$$

is residual. By construction, every $x \in R_0$ has dense orbit set under the countable family Σ .

Now define recursively

$$R_{m+1} := R_m \cap \bigcap_{h \in \Sigma} ((X \setminus \text{dom}(h)) \cup h^{-1}(R_m)), \quad m \geq 0.$$

Assume that R_m is residual. By Lemma 2.5, for each $h \in \Sigma$ the set

$$(X \setminus \text{dom}(h)) \cup h^{-1}(R_m)$$

is residual. Since Σ is countable, R_{m+1} is residual. Hence every R_m is residual, and so is

$$R_\infty := \bigcap_{m \geq 0} R_m.$$

If $x \in R_\infty \cap \text{dom}(h)$ for some $h \in \Sigma$, then for every m we have $x \in R_{m+1}$, hence $h(x) \in R_m$. Therefore $h(x) \in R_\infty$. In other words, R_∞ is forward invariant in the obvious partial sense.

Fix $x_0 \in R_\infty$. We construct inductively a sequence $x_n \in R_\infty$ and maps $h_n \in \Sigma$ with $x_n = h_n(x_{n-1}) \in V_n$. Indeed, since $x_{n-1} \in R_\infty \subset R_0$, the Σ -orbit set of x_{n-1} is dense, so there exists $h_n \in \Sigma$ with $x_{n-1} \in \text{dom}(h_n)$ and $h_n(x_{n-1}) \in V_n$. By forward invariance, $x_n \in R_\infty$.

For each $h \in \Sigma$, choose once and for all a finite representative word in the generators from $\mathcal{H}$. Concatenating the chosen words for $h_1, h_2, \dots$ yields an admissible infinite word in the original generators, starting at x_0 . Its orbit contains every x_n , hence meets every basis element V_n . Therefore that admissible branch is dense in X . $\square$

3 Global semigroup actions and dense-open restrictions

The next proposition is an extracted and slightly repackaged version of the finite-measure argument used by Koropec and Nassiri [13, §2]. We include a proof because it is repeatedly

used below. By a *forward branch* of the IFS generated by F we mean an orbit of the form

$$x, f_{i_1}(x), f_{i_2}f_{i_1}(x), f_{i_3}f_{i_2}f_{i_1}(x), \dots$$

along an infinite word $i_1, i_2, \dots$ in the generator set $\{1, \dots, m\}$.

Proposition 3.1 (Global finite-measure package). *Let X be a manifold, let μ be a finite Borel measure with full support on X , and let*

$$F = \{f_1, \dots, f_m\}$$

be a finite family of μ -preserving homeomorphisms of X . Write Γ^+ for the forward semigroup generated by F and Γ for the group generated by F . Then the following are equivalent:

- (i) Γ^+ is open-set transitive.
- (ii) There exists $x \in X$ such that $\Gamma^+(x)$ is dense in X .
- (iii) There exists a dense forward branch of the IFS generated by F .
- (iv) There exists a residual set $R' \subset X$ such that every $x \in R'$ admits a dense forward branch.

Proof. The implications (iv) $\Rightarrow$ (iii) $\Rightarrow$ (ii) are immediate. We prove (ii) $\Rightarrow$ (i) and (i) $\Rightarrow$ (iv).

Assume (ii). Then the Γ -orbit of the same point is dense. Hence the group action is open-set transitive: given nonempty open $U, V \subset X$, choose $g_1, g_2 \in \Gamma$ with $g_1(x) \in U$ and $g_2g_1(x) \in V$, so $g_2(U) \cap V \neq \emptyset$.

We claim that the semigroup action is open-set transitive as well. It suffices to prove the following statement: if $g \in \Gamma$ and $g(U) \cap V \neq \emptyset$ for nonempty open sets U, V , then there exists $\hat{g} \in \Gamma^+$ with $\hat{g}(U) \cap V \neq \emptyset$. We argue by induction on the length of a reduced word representing g .

If the reduced word has length one and $g \in F$, there is nothing to prove. If the reduced word has length one and $g^{-1} \in F$, let

$$W := g(U) \cap V.$$

Then W is a nonempty open set, hence $\mu(W) > 0$ by full support. Since g preserves μ , Poincaré recurrence gives a point $y \in W$ and arbitrarily large return times $n \geq 1$ with $g^n(y) \in W$; choose one with $n \geq 2$. Put $z := g^n(y)$. Since $z \in W \subset g(U)$, we can write $z = g(u)$ with $u \in U$. Therefore

$$y = g^{-n}(z) = g^{-(n-1)}(u) \in g^{-(n-1)}(U) \cap V.$$

Because $g^{-1} \in F$, the map $g^{-(n-1)}$ lies in Γ^+ , proving the base case.

Now let $g = g_{n+1} \cdots g_1$ be a reduced word of length $n+1$ with $g(U) \cap V \neq \emptyset$. Then

$$g_n \cdots g_1(U) \cap g_{n+1}^{-1}(V) \neq \emptyset.$$

By the induction hypothesis, there exists $h \in \Gamma^+$ such that

$$h(U) \cap g_{n+1}^{-1}(V) \neq \emptyset.$$

Applying the base case to the single letter g_{n+1} and the open set $h(U)$, we obtain $k \in \Gamma^+$ with $k(h(U)) \cap V \neq \emptyset$. Thus $kh \in \Gamma^+$ and $(kh)(U) \cap V \neq \emptyset$.

This proves (i). Now assume (i). Regard the countable set Γ^+ as a countable family of globally defined homeomorphisms and apply Theorem 2.6. We obtain a residual set of points admitting dense admissible branches for the family Γ^+ . Expanding each semigroup element in such a branch into a fixed representative word in the original generators from F gives a dense forward branch for the IFS generated by F . This proves (iv). $\square$

Remark 3.2. The implication (ii) $\Rightarrow$ (i) is exactly the finite-measure semigroup argument used in [13, §2]. The only additional input in Proposition 3.1 is the explicit forward-branch reformulation supplied by Theorem 2.6.

Corollary 3.3 (Dense-open restriction principle). *Let X be a second-countable Baire space, let $g_1, \dots, g_m$ be homeomorphisms of X whose forward semigroup is open-set transitive, and let $D_i \subset X$ be dense open subsets. Define partial homeomorphisms*

$$r_i := g_i|_{D_i} : D_i \rightarrow g_i(D_i), \quad i = 1, \dots, m.$$

Then the pseudosemigroup generated by $\{r_1, \dots, r_m\}$ is open-set transitive. Consequently, X contains a residual set of starting points admitting dense admissible branches for the restricted family.

Proof. Let $A, B \subset X$ be nonempty open sets. Choose a global word

$$w = g_{i_n} \circ \dots \circ g_{i_1}$$

such that $w(A) \cap B \neq \emptyset$. For $0 \leq k \leq n$ write

$$w_k := g_{i_k} \circ \dots \circ g_{i_1}, \quad w_0 := \text{id}_X.$$

Consider the admissibility set

$$\mathcal{A}_w := A \cap w^{-1}(B) \cap \bigcap_{k=0}^{n-1} w_k^{-1}(D_{i_{k+1}}).$$

The set $A \cap w^{-1}(B)$ is nonempty and open. Each $w_k^{-1}(D_{i_{k+1}})$ is dense open because w_k is a homeomorphism and $D_{i_{k+1}}$ is dense open. Hence $\mathcal{A}_w$ is nonempty and open. For every $x \in \mathcal{A}_w$, the restricted word $r_{i_n} \circ \dots \circ r_{i_1}$ is defined at x and equals $w(x) \in B$. This proves open-set transitivity for the restricted pseudosemigroup.

The residual dense-branch statement follows from Theorem 2.6. $\square$

4 Windowed graph-directed local systems and accessible cores

Let X be a second-countable Baire space. Fix a finite directed graph $G = (V, E)$ and an open decomposition

$$X = \bigsqcup_{v \in V} U_v$$

into nonempty atoms. We call each U_v an atom of the decomposition. All topological statements concerning subsets of a fixed atom U_v are understood relative to the topology of U_v .

Definition 4.1 (Windowed local system). A *windowed local system* over G consists, for each edge $e : v \rightarrow w$, of open sets

$$D_e \subset U_v, \quad I_e \subset U_w,$$

and a homeomorphism

$$f_e : D_e \rightarrow I_e.$$

Definition 4.2 (Admissible words and branches). An *admissible word* is a directed path $\alpha = e_n \cdots e_1$ in G . Its induced map is the maximal composition

$$f_\alpha = f_{e_n} \circ \cdots \circ f_{e_1}$$

defined on the subset of D_{e_1} for which each intermediate iterate lies in the domain of the next branch. An *admissible branch orbit* is a sequence $x_0, x_1, x_2, \dots \in X$ for which there exists an infinite directed path $e_1, e_2, \dots$ such that

$$x_n \in D_{e_{n+1}}, \quad x_{n+1} = f_{e_{n+1}}(x_n)$$

for all $n \geq 0$. The local system is *transitive* if some admissible branch orbit is dense in X .

Remark 4.3. The global branch relation on X is generally discontinuous as a multimap and need not extend across atom boundaries. This is why the partial-homeomorphism formalism from Section 2 is the natural one.

Example 4.4 (Hamiltonian windows). Let (M^4, Ω, H) be a Hamiltonian system and let $\Sigma \subset H^{-1}(h)$ be a two-dimensional transverse section. If $U_v \subset \Sigma$ are open atoms and the edges of G are realized by local Poincaré maps or scattering maps between windows $D_e \subset U_{s(e)}$ and $I_e \subset U_{t(e)}$, then Definition 4.1 applies whenever these transition maps are homeomorphisms. In the symplectic setting relevant later, the branches are in fact local area-preserving diffeomorphisms.

The next theorem is purely topological and is the deterministic propagation mechanism used later.

Theorem 4.5 (Residual dense branches from a partially accessible transitive core). *Let $\mathcal{F} = (D_e, I_e, f_e)_{e \in E}$ be a windowed local system over a finite directed graph $G = (V, E)$ on a second-countable Baire space $X = \bigsqcup_{v \in V} U_v$. Fix a distinguished vertex $v_* \in V$ and write $U_* := U_{v_*}$. Assume the following:*

- (a) *There exist two admissible loop words γ_0, γ_1 based at v_* whose induced partial homeomorphisms*

$$R_0 : D_0 \rightarrow I_0, \quad R_1 : D_1 \rightarrow I_1,$$

have dense open domains and images in U_ , and whose generated pseudosemigroup is open-set transitive on U_* .*

- (b) *For every vertex $v \in V$ there exist admissible words*

$$a_v : v_* \rightsquigarrow v, \quad b_v : v \rightsquigarrow v_*,$$

whose induced maps are partial homeomorphisms

$$A_v : P_v \rightarrow Q_v \subset U_v, \quad B_v : E_v \rightarrow F_v \subset U_*,$$

with $P_v, F_v \subset U_$ dense open in U_* and $Q_v, E_v \subset U_v$ dense open in U_v .*

Then the pseudosemigroup of all admissible words of the local system is open-set transitive on X . Consequently, for every vertex $v \in V$, the atom U_v contains a residual subset of starting points that admit dense admissible branches in X .

Proof. Let $A, B \subset X$ be nonempty open sets. Since the atoms form an open partition, after shrinking we may assume that $A \subset U_u$ and $B \subset U_v$ for some vertices $u, v \in V$.

Because $E_u \subset U_u$ is dense open in U_u , the set $A_0 := A \cap E_u$ is nonempty open in U_u . Applying the partial homeomorphism $B_u : E_u \rightarrow F_u$, we obtain a nonempty open set $B_u(A_0) \subset U_*$. Likewise, since $Q_v \subset U_v$ is dense open in U_v , the set $B \cap Q_v$ is nonempty open in U_v , so $A_v^{-1}(B \cap Q_v)$ is a nonempty open subset of U_* .

By hypothesis (a), the pseudosemigroup generated by R_0 and R_1 is open-set transitive on U_* . Hence there exists a core word C such that

$$C(B_u(A_0) \cap \text{dom}(C)) \cap A_v^{-1}(B \cap Q_v) \neq \emptyset.$$

Choose $x \in A_0$ such that $B_u(x) \in \text{dom}(C)$ and $A_v(C(B_u(x))) \in B$. Then the concatenated admissible word $a_v C b_u$ is defined at $x \in A$ and lands in B . Thus the full pseudosemigroup is open-set transitive on X .

There are only countably many admissible words in the finite graph G , and each induces a partial homeomorphism of X . Applying Theorem 2.6 to this countable family yields a residual set $R_\infty \subset X$ such that every $x \in R_\infty$ admits a dense admissible branch. Since each U_v is open in X , the intersection $R_\infty \cap U_v$ is residual in U_v . $\square$

Remark 4.6. Theorem 4.5 is the deterministic core-to-network propagation mechanism needed later. No smoothness, measure preservation, or symplectic structure is used in the proof.

5 The universal dense-open window theorem for generic conservative pairs

Theorem 5.1 (Koropecski–Nassiri). *Let (S, ω) be a closed orientable surface and let $r \in \mathbb{N} \cup \{\infty\}$. There exists a residual subset*

$$\mathcal{R}_{\text{KN}} \subset \text{Diff}_\omega^r(S) \times \text{Diff}_\omega^r(S)$$

with the product C^r topology such that for every $(f, g) \in \mathcal{R}_{\text{KN}}$, the iterated function system generated by f and g is transitive on S .

Remark 5.2. The erratum [14] corrects part of the higher-genus fiberwise discussion in [13]. Since the present paper uses only the pair theorem stated above, the correction affects our work only at the level of attribution.

Theorem 5.3 (Universal dense-open window theorem for conservative pairs). *Let (S, ω) be a closed orientable surface and let $r \in \mathbb{N} \cup \{\infty\}$. There exists a residual subset*

$$\mathcal{R}_{\text{univ}} \subset \text{Diff}_\omega^r(S) \times \text{Diff}_\omega^r(S)$$

with the following property: for every $(g_0, g_1) \in \mathcal{R}_{\text{univ}}$ and for every pair of dense open sets $D_0, D_1 \subset S$, the restricted partial homeomorphisms

$$r_0 := g_0|_{D_0} : D_0 \rightarrow g_0(D_0), \quad r_1 := g_1|_{D_1} : D_1 \rightarrow g_1(D_1)$$

generate an open-set transitive pseudosemigroup on S . Consequently, for every such choice of D_0, D_1 , the surface S contains a residual subset of points admitting dense admissible branches for the restricted pair (r_0, r_1) .

Proof. Set $\mathcal{R}_{\text{univ}} := \mathcal{R}_{\text{KN}}$. Fix $(g_0, g_1) \in \mathcal{R}_{\text{KN}}$ and let $D_0, D_1 \subset S$ be arbitrary dense open subsets. By Theorem 5.1, the IFS generated by g_0 and g_1 is transitive on S . By Proposition 3.1, the global forward semigroup generated by g_0 and g_1 is open-set transitive. Corollary 3.3 now applies to the dense open restrictions to D_0 and D_1 . $\square$

Remark 5.4. Theorem 5.3 is the main bridge from generic global conservative transitivity to local window dynamics. The key point is the *universality*: the residual set of global pairs does not depend on the windows.

6 Fibered genericity, two-edge forcing, and a universal private-edge theorem

We now pass from a single surface to a finite graph-directed family of copies of the same surface:

$$M = S \times V = \bigsqcup_{v \in V} S_v, \quad S_v := S \times \{v\}.$$

Definition 6.1 (Conservative graph-directed IFS). Let (S, ω) be a closed orientable surface and let $G = (V, E)$ be a finite directed graph. A *conservative graph-directed IFS of class C^r over G* is an assignment

$$\mathcal{F} = (f_e)_{e \in E}, \quad f_e \in \text{Diff}_\omega^r(S),$$

which acts on M by

$$F_e(x, s(e)) = (f_e(x), t(e)).$$

Definition 6.2 (Two-loop scheme with private single-use edges). A finite directed graph G admits a *two-loop scheme with private single-use edges* at a vertex v_* if there exist two based admissible loop words

$$\gamma_0, \gamma_1$$

at v_* and edges

$$\varepsilon_0, \varepsilon_1 \in E$$

such that, for each $i \in \{0, 1\}$,

- (i) the edge ε_i occurs exactly once in γ_i ,
- (ii) the edge ε_{1-i} does not occur in γ_i .

The loop words need not be simple and may overlap arbitrarily outside the two distinguished private edges.

Example 6.3 (A three-section chain with a shared heteroclinic channel). Consider the directed graph G with vertex set $V = \{v_*, 1, 2\}$ and edge set

$$E = \{a, b, c, \varepsilon_0, \varepsilon_1\},$$

where the edges are

$$a : v_* \rightarrow 1, \quad b : 1 \rightarrow 2, \quad c : 2 \rightarrow v_*, \quad \varepsilon_0 : v_* \rightarrow 1, \quad \varepsilon_1 : 2 \rightarrow v_*.$$

Define two based loop words at v_* :

$$\gamma_0 = c b \varepsilon_0 : v_* \xrightarrow{\varepsilon_0} 1 \xrightarrow{b} 2 \xrightarrow{c} v_*, \quad \gamma_1 = \varepsilon_1 b a : v_* \xrightarrow{a} 1 \xrightarrow{b} 2 \xrightarrow{\varepsilon_1} v_*.$$

The two loops *share* the middle edge $b : 1 \rightarrow 2$, so they are not edge-disjoint. Nonetheless, the pair $(\varepsilon_0, \varepsilon_1)$ satisfies the private single-use condition of Definition 6.2:

- ε_0 occurs exactly once in γ_0 and does not occur in γ_1 ;
- ε_1 occurs exactly once in γ_1 and does not occur in γ_0 .

The return maps at the core vertex v_* are

$$R_0 = f_c \circ f_b \circ f_{\varepsilon_0}, \quad R_1 = f_{\varepsilon_1} \circ f_b \circ f_a.$$

If we fix the shared edge map f_b and the transport maps f_a and f_c , then R_0 depends only on f_{ε_0} and R_1 depends only on f_{ε_1} , via

$$R_0 = (f_c \circ f_b) \circ f_{\varepsilon_0}, \quad R_1 = f_{\varepsilon_1} \circ (f_b \circ f_a).$$

Thus the return-map pair (R_0, R_1) depends on the private-edge pair $(f_{\varepsilon_0}, f_{\varepsilon_1})$ through independent left and right translations. By Corollary 6.6, there is a residual set of choices of $(f_{\varepsilon_0}, f_{\varepsilon_1})$ for which the resulting graph-directed system admits dense admissible branches from residually many starting points in every component—and this holds for every pair of dense open windows $D_0, D_1 \subset S$ simultaneously.

To see concretely why the private-edge hypothesis is needed and why edge-disjointness is too strong, observe the following. With five edges and three vertices, any loop based at v_* that visits both satellite vertices must traverse $b : 1 \rightarrow 2$; there is no alternative path from vertex 1 to vertex 2. Hence *every* pair of non-self-loop based loops at v_* that visit both satellites shares the edge b , and the edge-disjoint hypothesis fails for every such pair. Yet the private-edge condition is easily satisfied, precisely because the two loops can enter and exit the shared channel through different edges.

In the Hamiltonian interpretation, the three vertices represent transverse sections $\Sigma_*, \Sigma_1, \Sigma_2$ in an energy surface, the shared edge b corresponds to a unique heteroclinic channel connecting sections 1 and 2 (for instance, a branch of the stable/unstable manifold of a normally hyperbolic invariant manifold), while ε_0 and ε_1 represent distinct scattering maps at the entry and exit of the channel. The theorem guarantees that generic choices of these two scattering maps suffice to force dense orbits through the entire network.

Lemma 6.4 (Sectionwise residual criterion). *Let X and Y be Polish spaces, and let $A \subset X \times Y$ have the Baire property. If every section*

$$A_x := \{y \in Y : (x, y) \in A\}$$

is residual in Y , then A is residual in $X \times Y$.

Proof. This is a standard consequence of the Kuratowski–Ulam theorem; see, for instance, [12, Thm. 8.41]. $\square$

Theorem 6.5 (Universal fibered genericity with fixed off-loop data and private single-use edges). *Let (S, ω) be a closed orientable surface, let $r \in \mathbb{N} \cup \{\infty\}$, and let $G = (V, E)$ be a finite strongly connected directed graph. Assume that G admits a two-loop scheme with private single-use edges at a vertex v_* , witnessed by based loop words γ_0, γ_1 and private edges $\varepsilon_0, \varepsilon_1$. Let*

$$E_{\text{loop}} := E(\gamma_0) \cup E(\gamma_1), \quad E_{\text{off}} := E \setminus E_{\text{loop}}.$$

Fix an arbitrary off-loop assignment

$$\mathbf{h} = (h_e)_{e \in E_{\text{off}}} \in \prod_{e \in E_{\text{off}}} \text{Diff}_{\omega}^r(S).$$

Then there exists a residual subset

$$\mathcal{R}_G(\mathbf{h}) \subset \prod_{e \in E_{\text{loop}}} \text{Diff}_{\omega}^r(S)$$

with the following property: for every on-loop assignment $\mathbf{g} \in \mathcal{R}_G(\mathbf{h})$ and for every pair of dense open subsets $D_0, D_1 \subset S$, each component S_v contains a residual subset of starting points admitting dense admissible branches in M after passing to the derived macro-edge system in which the two based loop words γ_0, γ_1 are used as core return maps and then restricted to D_0 and D_1 , respectively.

Proof. Let

$$H := \text{Diff}_\omega^r(S), \quad \mathcal{P}_{\text{loop}} := \prod_{e \in E_{\text{loop}}} H, \quad Y := \prod_{e \in E_{\text{loop}} \setminus \{\varepsilon_0, \varepsilon_1\}} H.$$

Since $\mathcal{R}_{\text{KN}}$ is residual in $H \times H$, replace it by a dense G_δ subset if necessary and still denote it by $\mathcal{R}_{\text{KN}}$.

For each $\eta \in Y$ and each pair $(u_0, u_1) \in H \times H$, form the on-loop assignment obtained by inserting u_i at the edge ε_i and the coordinates of η on all other edges in E_{loop} . Because ε_i occurs exactly once in γ_i and ε_{1-i} does not occur in γ_i , the return maps along the chosen loop words have the form

$$R_0(\eta, u_0) = L_0(\eta) \circ u_0 \circ M_0(\eta), \quad R_1(\eta, u_1) = L_1(\eta) \circ u_1 \circ M_1(\eta),$$

for some $L_i(\eta), M_i(\eta) \in H$ depending continuously on η . Define

$$\Theta_\eta : H \times H \rightarrow H \times H, \quad \Theta_\eta(u_0, u_1) := (R_0(\eta, u_0), R_1(\eta, u_1)).$$

Each Θ_η is a homeomorphism, since it is a product of left and right translations in the topological group H .

Now let $A \subset Y \times H \times H \cong \mathcal{P}_{\text{loop}}$ be the set of all on-loop assignments for which the pair of return maps belongs to $\mathcal{R}_{\text{KN}}$. Equivalently,

$$A := \{(\eta, u_0, u_1) : \Theta_\eta(u_0, u_1) \in \mathcal{R}_{\text{KN}}\}.$$

Because the return-map assignment is continuous and $\mathcal{R}_{\text{KN}}$ is G_δ , the set A is a G_δ subset of $Y \times H \times H$. For every fixed $\eta \in Y$, the section

$$A_\eta = \Theta_\eta^{-1}(\mathcal{R}_{\text{KN}})$$

is residual in $H \times H$. Therefore Lemma 6.4 implies that A is residual in $\mathcal{P}_{\text{loop}}$. Set

$$\mathcal{R}_G(\mathbf{h}) := A.$$

Fix $\mathbf{g} \in \mathcal{R}_G(\mathbf{h})$ and let $D_0, D_1 \subset S$ be arbitrary dense open subsets. Then the two return maps along γ_0 and γ_1 form a pair in $\mathcal{R}_{\text{KN}}$. By Theorem 5.3, the restricted return maps

$$R_0|_{D_0}, \quad R_1|_{D_1}$$

generate an open-set transitive pseudosemigroup on the core component $S_{v_*} \cong S$.

Because G is strongly connected, for each vertex $v \in V$ choose once and for all a path $a_v : v_* \rightsquigarrow v$ and a path $b_v : v \rightsquigarrow v_*$. Since every edge map in the completed graph-directed system is globally defined on the relevant copy of S , these paths induce global homeomorphisms

$$A_v : S_{v_*} \rightarrow S_v, \quad B_v : S_v \rightarrow S_{v_*}.$$

Thus the hypotheses of Theorem 4.5 are satisfied for the derived macro-edge system consisting of the two restricted core return maps and the transport maps A_v, B_v . The conclusion of Theorem 4.5 gives a residual set of dense admissible branches in every component S_v for the macro-edge system.

Finally, expanding each macro-edge into its fixed representative path in the original graph gives dense admissible branches for the original graph-directed system. $\square$

Corollary 6.6 (Two-edge forcing, universal window version). *Under the graph-theoretic hypotheses of Theorem 6.5, fix arbitrary conservative assignments for every edge of G except the two private edges ε_0 and ε_1 . Then there exists a residual subset*

$$\mathcal{R}_{\varepsilon_0, \varepsilon_1} \subset \text{Diff}_\omega^r(S) \times \text{Diff}_\omega^r(S)$$

with the following property: for every choice of edge maps on $(\varepsilon_0, \varepsilon_1)$ from this set and for every pair of dense open subsets $D_0, D_1 \subset S$, the resulting graph-directed system on $M = S \times V$ has residual dense-branch sets in every component for the corresponding derived macro-edge system with core loop returns restricted to D_0 and D_1 .

Proof. Fix all edges except ε_0 and ε_1 . Then the return maps along γ_0 and γ_1 have the form

$$R_0(g_0) = L_0 \circ g_0 \circ M_0, \quad R_1(g_1) = L_1 \circ g_1 \circ M_1,$$

for fixed conservative diffeomorphisms $L_i, M_i \in \text{Diff}_\omega^r(S)$. Hence

$$\Theta : \text{Diff}_\omega^r(S) \times \text{Diff}_\omega^r(S) \rightarrow \text{Diff}_\omega^r(S) \times \text{Diff}_\omega^r(S), \quad \Theta(g_0, g_1) := (L_0 \circ g_0 \circ M_0, L_1 \circ g_1 \circ M_1)$$

is a homeomorphism. Set

$$\mathcal{R}_{\varepsilon_0, \varepsilon_1} := \Theta^{-1}(\mathcal{R}_{\text{KN}}).$$

This set is residual. For every pair (g_0, g_1) in it, the return pair belongs to $\mathcal{R}_{\text{KN}}$, so the conclusion follows from Theorem 6.5. $\square$

Corollary 6.7 (Global residual theorem in the full parameter space). *Under the graph-theoretic hypotheses of Theorem 6.5, there exists a residual subset*

$$\tilde{\mathcal{R}}_G \subset \prod_{e \in E} \text{Diff}_\omega^r(S)$$

with the product C^r topology such that for every complete assignment $\mathcal{F} \in \tilde{\mathcal{R}}_G$ and every pair of dense open subsets $D_0, D_1 \subset S$, each component S_v contains a residual set of points admitting dense admissible branches in M for the corresponding derived macro-edge system with the two chosen core loop returns restricted to D_0 and D_1 .

Proof. Let

$$H := \text{Diff}_\omega^r(S), \quad Y := \prod_{e \in E \setminus \{\varepsilon_0, \varepsilon_1\}} H.$$

For each fixed $\eta \in Y$, the pair of return maps depends on the two distinguished coordinates $(u_0, u_1) \in H \times H$ by a homeomorphism

$$\Theta_\eta(u_0, u_1) = (L_0(\eta) \circ u_0 \circ M_0(\eta), L_1(\eta) \circ u_1 \circ M_1(\eta)).$$

Let $A \subset Y \times H \times H \cong \prod_{e \in E} H$ be the set of complete assignments for which the return pair belongs to $\mathcal{R}_{\text{KN}}$. Exactly as before, A has the Baire property and every section A_η is residual in $H \times H$. By Lemma 6.4, the set A is residual in the full parameter space. Set

$$\tilde{\mathcal{R}}_G := A.$$

For every complete assignment in this residual set, the two core return maps form a pair in $\mathcal{R}_{\text{KN}}$, so the conclusion follows from Theorem 6.5. $\square$

Remark 6.8. The private-edge argument above is stronger than the usual edge-disjoint parameter step in two ways. First, the two loop words may overlap arbitrarily away from the distinguished edges. Second, the residual parameter set is universal in the core windows D_0, D_1 .

7 A conditional Hamiltonian corollary

The preceding results were designed to fit the return–scattering formalism from Hamiltonian dynamics, but the conclusion should be read as a conditional transfer principle rather than as a generic theorem for a fixed Hamiltonian family.

Corollary 7.1 (Conditional Hamiltonian window network theorem). *Let (M^4, Ω, H) be a C^r Hamiltonian system, let $\mathcal{E} = H^{-1}(h)$ be a regular energy surface, and let*

$$\Sigma_*, \Sigma_v \subset \mathcal{E}$$

be finitely many embedded two-dimensional transverse sections. Assume that the distinguished core section Σ_ is a closed orientable surface, endowed with the induced area form $\omega_* := \Omega|_{\Sigma_*}$, and write $U_* := \Sigma_*$. For each non-core vertex v let $U_v \subset \Sigma_v$ be a nonempty open atom. Suppose that the non-core transport edges of a finite directed graph $G = (V, E)$ are realized by fixed C^r symplectic local transition maps*

$$f_e : D_e \rightarrow I_e$$

between open windows $D_e \subset U_{s(e)}$ and $I_e \subset U_{t(e)}$, realized either by local Poincaré maps or by scattering maps, and that these transport words satisfy hypothesis (b) of Theorem 4.5

after collapsing two chosen core loop templates to macro-edges based at the distinguished vertex v_* .

Then there exists a residual set

$$\mathcal{R}_* \subset \text{Diff}_{\omega_*}^r(\Sigma_*) \times \text{Diff}_{\omega_*}^r(\Sigma_*)$$

with the following property: for every pair $(g_0, g_1) \in \mathcal{R}_*$ and every dense open windows $D_0, D_1 \subset \Sigma_*$, the window network obtained by using $g_0|_{D_0}$ and $g_1|_{D_1}$ as the two core macro-transitions and keeping all transport maps fixed has the property that every atom U_v contains a residual set of points admitting dense admissible itineraries through the network.

Proof. Take $\mathcal{R}_* := \mathcal{R}_{\text{univ}}$ from Theorem 5.3. For every pair $(g_0, g_1) \in \mathcal{R}_*$ and every dense open $D_0, D_1 \subset \Sigma_*$, the restricted core maps $g_0|_{D_0}$ and $g_1|_{D_1}$ generate an open-set transitive pseudosemigroup on Σ_* . The fixed transport data satisfy hypothesis (b) of Theorem 4.5, so that theorem applies to the resulting macro-edge system. $\square$

Remark 7.2. Corollary 7.1 does not assert that a given Hamiltonian family realizes a residual set of global core return pairs. Its content is conditional and topological: once two core transitions are modeled by a generic conservative pair on a closed section, dense-open window transitivity propagates through the whole return-scattering network.

Remark 7.3 (Toward an unconditional Hamiltonian statement). The gap between Corollary 7.1 and a fully unconditional generic Hamiltonian theorem is the following. In a concrete Hamiltonian family H_ε , the Poincaré return maps to a transverse section are *determined* by the Hamiltonian and the energy surface; they are not freely chosen from

$$\text{Diff}_{\omega_*}^r(\Sigma_*) \times \text{Diff}_{\omega_*}^r(\Sigma_*).$$

Thus one cannot pass directly from the residual subset $\mathcal{R}_{\text{KN}}$ of conservative pairs to a residual subset of Hamiltonian parameters without additional information on the parameter-to-return-map correspondence.

More precisely, let $\mathcal{P}$ be a Baire space of admissible perturbation parameters and suppose that to each $\varepsilon \in \mathcal{P}$ there is associated a pair of induced core return maps

$$\Phi(\varepsilon) := (R_0(\varepsilon), R_1(\varepsilon)) \in \text{Diff}_{\omega_*}^r(\Sigma_*) \times \text{Diff}_{\omega_*}^r(\Sigma_*).$$

To deduce a generic-parameter statement from Theorem 5.3, it is not enough to know merely that the image of Φ meets $\mathcal{R}_{\text{KN}}$, or even that the image is non-meager. What is needed is a mechanism ensuring that the preimage

$$\Phi^{-1}(\mathcal{R}_{\text{KN}})$$

is residual in $\mathcal{P}$. A sufficient condition would be that Φ be continuous and category-preserving; for example, it would suffice that Φ be continuous and open onto a nonempty open subset of its image space. Under such a hypothesis, the residual set $\mathcal{R}_{\text{KN}}$ would pull

back to a residual set of parameters, and Corollary 7.1 would then yield dense admissible itineraries for Baire-generic perturbations in the family.

In frameworks such as Delshams–de la Llave–Seara [6, 7], one may hope to obtain this kind of conclusion by proving sufficiently strong transversality or local-surjectivity properties for the map Φ , using the freedom to perturb the Hamiltonian near the homoclinic channel and thereby vary the induced scattering and return dynamics. Establishing such properties is a substantial analytic problem that depends on the specific Hamiltonian family and lies beyond the purely topological methods of the present paper.

8 Numerical illustrations

We present four experiments on $\mathbb{T}^2 = [0, 1)^2$ with the standard area form. The first two illustrate the main theorems; the third and fourth probe the sharpness of the dense-open and genericity hypotheses, respectively. All experiments use the area-preserving shear maps

$$g_0(x, y) = (x + 0.70 \sin(2\pi y), y) \pmod{1}, \quad g_1(x, y) = (x, y + 0.63 \sin(2\pi x)) \pmod{1}.$$

Each map is the time-one map of a smooth Hamiltonian shear and therefore preserves area. Individually, g_0 preserves the y -coordinate and g_1 preserves the x -coordinate, so neither is topologically transitive by itself.

8.1 Experiment 1: a dense-open window pair on the torus

This experiment illustrates Theorem 5.3. We define dense-open windows

$$D_0 = \mathbb{T}^2 \setminus (C_{0,1} \cup C_{0,2}), \quad D_1 = \mathbb{T}^2 \setminus (C_{1,1} \cup C_{1,2}),$$

where $C_{0,1}, C_{0,2}$ are two smooth vertical-type curves and $C_{1,1}, C_{1,2}$ are two smooth horizontal-type curves. Starting from $(\sqrt{2} - 1, \sqrt{3} - 1)$ and applying a random admissible branch of length 1.2×10^5 , we observe rapid coverage of the torus. Figure 1 shows the orbit together with the removed curves and the growth of visited grid cells: the branch visits more than 99.9% of the 120×120 grid.

8.2 Experiment 2: a two-state return–scattering toy network

This experiment illustrates Theorem 6.5 and the return–scattering interpretation behind Corollary 7.1. Consider the graph with two vertices 0, 1, two loops at the core vertex 0, and transport edges $p : 0 \rightarrow 1$ and $q : 1 \rightarrow 0$. On the core section $S_0 \cong \mathbb{T}^2$ we use the two shear maps above. On the transport edges we use the area-preserving translations

$$p(x, y) = (x + 0.137, y + 0.211) \pmod{1}, \quad q(x, y) = (x + 0.318, y + 0.127) \pmod{1}.$$

Starting from a point in S_1 , we follow an admissible branch that chooses a core loop with probability 0.8 when in S_0 and uses the transport edge p with probability 0.2; whenever

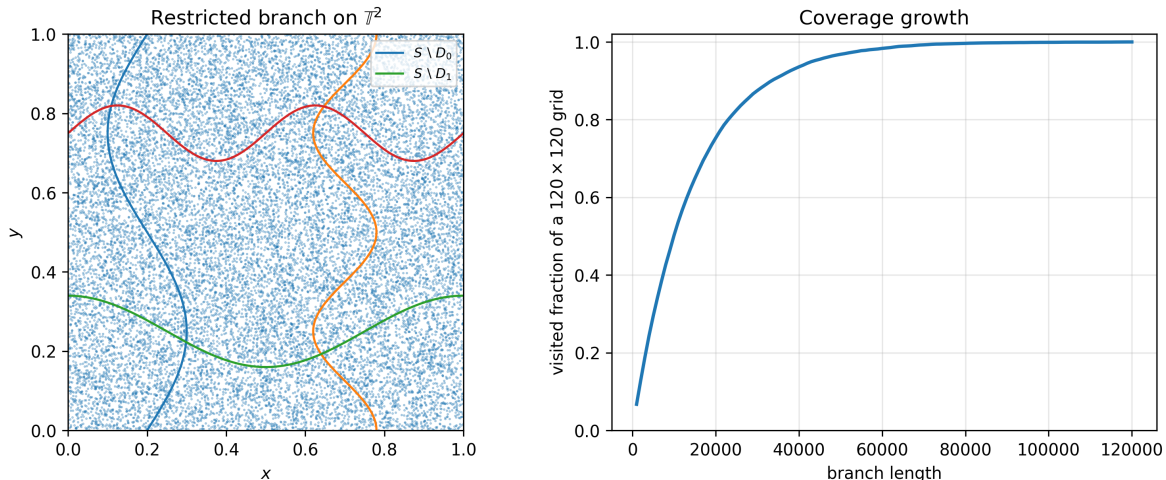


Figure 1: Illustration of Theorem 5.3. Left: a random admissible branch for the restricted pair $(g_0|_{D_0}, g_1|_{D_1})$ on $\mathbb{T}^2$, with the removed dense-open window boundaries overlaid. Right: fraction of a 120×120 grid visited by the branch as a function of branch length.

the orbit reaches S_1 , it immediately returns via q . Figure 4 shows the resulting coverage in both components: the branch visits essentially all grid cells in S_0 and more than 95% of the grid cells in S_1 .

8.3 Experiment 3: sharpness of the dense-open hypothesis

The dense-open condition on the windows is used in an essential way in Corollary 3.3 and Theorem 5.3. To probe its necessity, we replace the measure-zero curve boundaries of Experiment 1 with *fat strips* of positive area: $D_0 = \mathbb{T}^2 \setminus \{0.35 < y < 0.65\}$ and $D_1 = \mathbb{T}^2 \setminus \{0.35 < x < 0.65\}$. These windows are open but their complements are *not* nowhere dense, so the dense-open hypothesis fails.

Figure 2 compares the resulting branch with the dense-open branch from Experiment 1 over the same number of iterates. With the fat-strip windows the orbit is visibly confined: the coverage curve plateaus near 91%, leaving significant regions of the torus permanently unvisited. By contrast, the dense-open branch reaches 100% coverage. This illustrates that the dense-open assumption is not merely a technical convenience: when the removed set has positive measure and disconnects portions of the phase space, the restriction principle breaks down and the admissible branch can fail to be dense.

8.4 Experiment 4: sharpness of the genericity hypothesis on private edges

Corollary 6.6 asserts that a residual set of choices for the two private-edge maps suffices to force dense branching through the network. To probe the necessity of genericity, we return

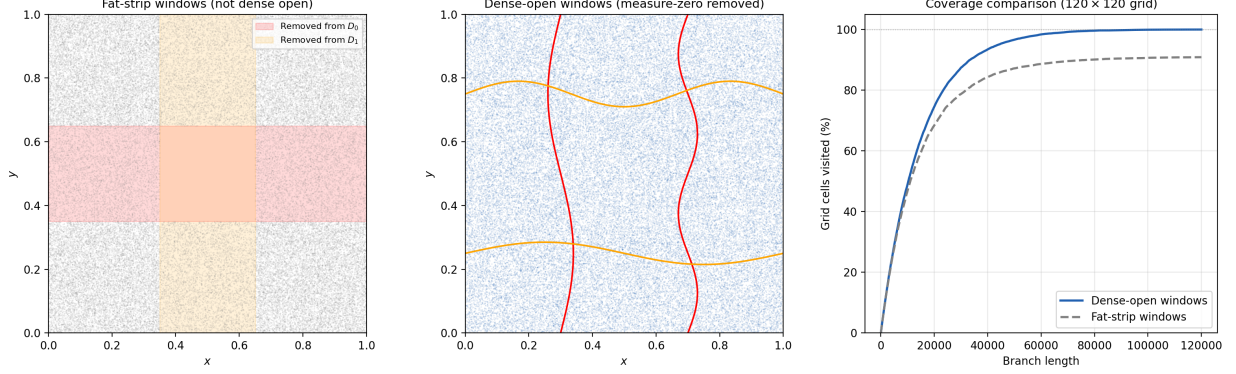


Figure 2: Sharpness of the dense-open hypothesis. Left: admissible branch with fat-strip windows (forbidden regions shaded). Center: admissible branch with dense-open (measure-zero) windows. Right: coverage curves showing that the fat-strip branch plateaus while the dense-open branch reaches full coverage.

to the three-section graph of Example 6.3 and compare two scenarios on the core section $S_{v_*} \cong \mathbb{T}^2$. In both cases the shared edge map f_b and the transport maps f_a, f_c are fixed as *horizontal* irrational translations (preserving the y -coordinate). In the *non-generic* scenario, the two private-edge maps f_{ε_0} and f_{ε_1} are set to the identity; the resulting return maps $R_0 = f_c \circ f_b$ and $R_1 = f_b \circ f_a$ are also horizontal translations, and every orbit is confined to a single horizontal circle. In the *generic* scenario, $f_{\varepsilon_0} = g_0$ and $f_{\varepsilon_1} = g_1$ are the shear maps used above, which couple the two coordinates.

Figure 3 shows the resulting core orbits. With identity private edges, the orbit traces a single horizontal line and coverage stalls at approximately 1% of the grid. With generic shear maps on the private edges, the orbit rapidly fills the torus and coverage reaches 100%. This confirms that the genericity condition on the private edges is essential: degenerate (e.g. identity) choices can confine orbits to invariant submanifolds, and it is the generic breaking of these invariances that forces transitivity.

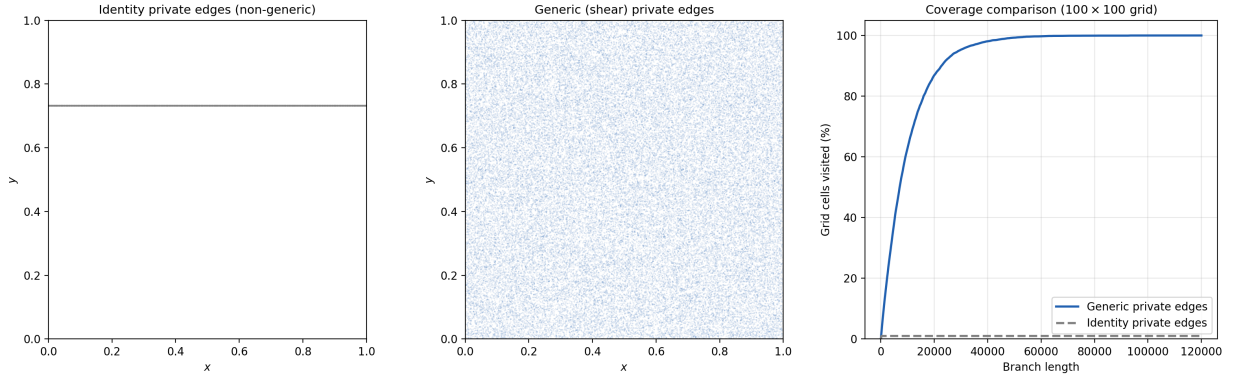


Figure 3: Sharpness of the genericity hypothesis on private edges. Left: core orbit with identity private-edge maps (non-generic), confined to a single horizontal circle ($\approx 1\%$ coverage). Center: core orbit with generic (shear) private-edge maps (100% coverage). Right: coverage comparison confirming that genericity in the private edges is essential.

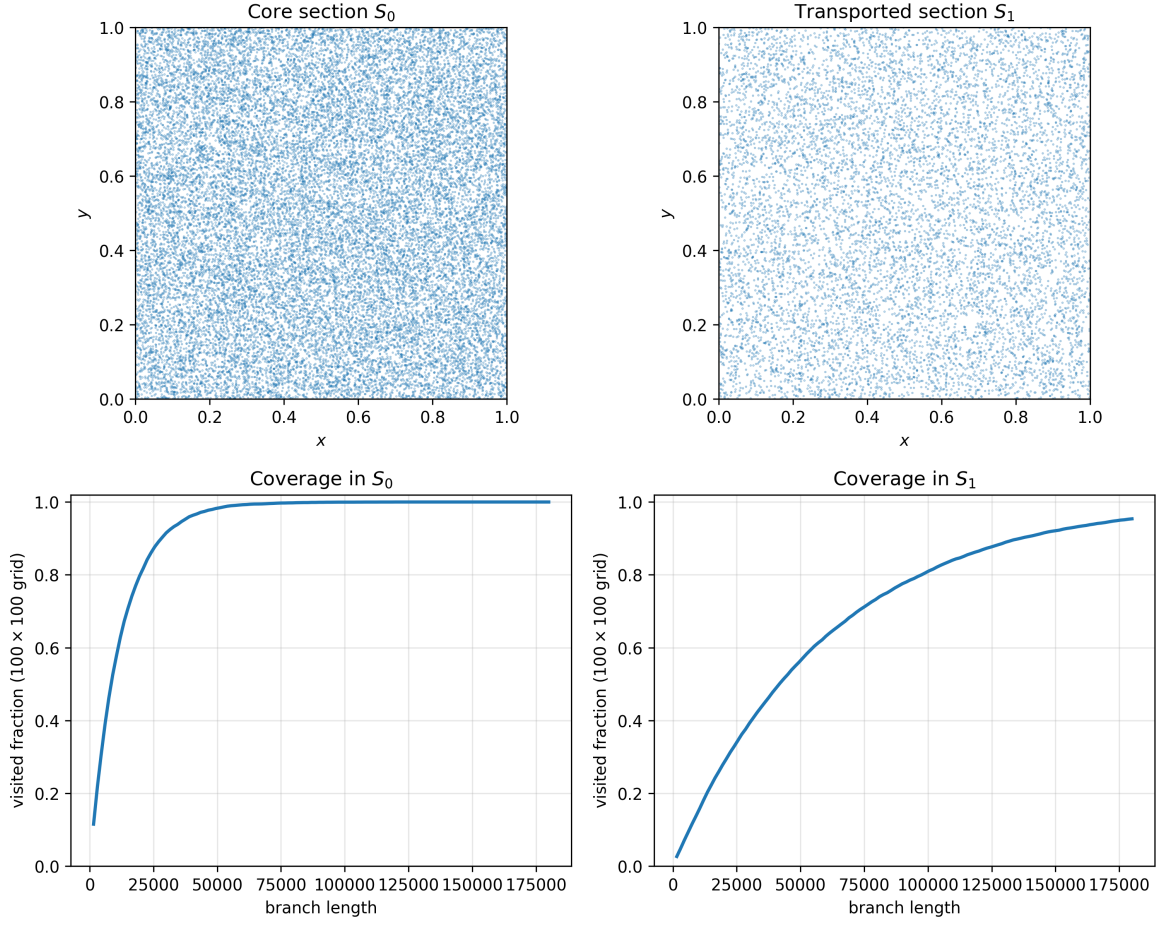


Figure 4: Heuristic illustration of Theorem 6.5 and the Hamiltonian network viewpoint. Top row: orbit samples on the core section S_0 and the transported section S_1 . Bottom row: coverage curves on 100×100 grids. In this simulation the branch visits essentially all grid cells in S_0 and more than 95% of the grid cells in S_1 .

9 Discussion

Summary of contributions

The paper establishes a chain of results that reduces the local windowed transitivity problem to a single external input—the Koropecski–Nassiri generic transitivity theorem for conservative pairs on a closed surface. The universal window theorem (Theorem 5.3) shows that a single residual set of area-preserving pairs works for *all* dense open windows simultaneously, resolving an apparent uncountable-intersection obstacle: the universal quantifier over windows is absorbed inside the dense-open restriction principle (Corollary 3.3) rather than at the level of the residual set. The core-to-network propagation theorem (Theorem 4.5) then transfers this local core transitivity through genuinely partial window networks by a purely topological argument. Finally, the private-edge parameter theorem (Theorem 6.5) and its corollaries show that genericity in just two edge parameters—which need not lie on edge-disjoint loops—suffices to force dense admissible branches throughout a graph-directed conservative system, with all other edge assignments held fixed and all dense open windows handled universally.

The dense-open boundary

The density of the windows is used in an essential way. In the proof of Corollary 3.3, the admissibility set for a global word w of length n is obtained by intersecting a nonempty open set with n dense open pullbacks; the Baire property guarantees this intersection is nonempty. When the windows have nonempty interior in their complements, these pullbacks need not be dense, and the intersection can collapse. The numerical experiments of Section 8 confirm this: fat-strip windows with positive-measure complements produce branches that visibly fail to cover the phase space (Figure 2), while measure-zero curve boundaries yield full coverage (Figure 1). Any extension of the present results to windows with positive-codimension boundaries but positive-measure complements would require genuinely new geometric input, likely involving quantitative recurrence estimates or controlled distortion bounds for compositions landing in the allowed region.

Closed versus open surfaces

The Koropecski–Nassiri theorem, and hence Theorem 5.3, applies to *closed* orientable surfaces. In many Hamiltonian applications, however, the relevant Poincaré section is an open disk or annulus with finite area, not a closed surface. The topological machinery of Sections 2 and 4—the forward-branch Birkhoff theorem, the dense-open restriction principle, and the core-to-network propagation—uses only the Baire property and second countability, and therefore applies equally well to open surfaces. The only obstruction to extending the full package to the open-surface setting is the absence of a Koropecski–Nassiri type theorem: one would need to establish that generic pairs of area-preserving diffeomorphisms of an open

finite-area surface generate a transitive semigroup. If such a result were proved—whether for the open disk, the annulus, or a punctured surface—the universal window theorem and all its graph-directed consequences would extend immediately, with no changes to the arguments of the present paper.

Toward unconditional Hamiltonian applications

As discussed in Remark 7.3, the Hamiltonian corollary (Corollary 7.1) is conditional: it assumes that the core return maps are modeled by a freely chosen generic conservative pair, whereas in a concrete Hamiltonian family the return maps are determined by the Hamiltonian and the energy surface. Thus one cannot pass directly from the residual subset $\mathcal{R}_{\text{KN}}$ of

$$\text{Diff}_{\omega_*}^r(\Sigma_*) \times \text{Diff}_{\omega_*}^r(\Sigma_*)$$

to a residual subset of Hamiltonian parameters without additional information on the parameter-to-return-map correspondence.

More precisely, let $\mathcal{P}$ be a Baire space of admissible perturbation parameters and suppose that to each $\varepsilon \in \mathcal{P}$ there is associated a pair of induced core return maps

$$\Phi(\varepsilon) := (R_0(\varepsilon), R_1(\varepsilon)) \in \text{Diff}_{\omega_*}^r(\Sigma_*) \times \text{Diff}_{\omega_*}^r(\Sigma_*).$$

To obtain a generic-parameter consequence from Theorem 5.3, it is not enough to know merely that the image of Φ meets $\mathcal{R}_{\text{KN}}$, or even that the image is non-meager. What is needed is a mechanism ensuring that the preimage

$$\Phi^{-1}(\mathcal{R}_{\text{KN}})$$

is residual in $\mathcal{P}$. A sufficient condition would be that Φ be continuous and category-preserving; for example, it would suffice that Φ be continuous and open onto a nonempty open subset of its target (or more generally, of its image). Under such a hypothesis, the residual set $\mathcal{R}_{\text{KN}}$ would pull back to a residual set of Hamiltonian parameters, and Corollary 7.1 would then yield dense admissible itineraries for Baire-generic perturbations in the family.

The most promising setting for such a program is the framework of Delshams–de la Llave–Seara [6, 7], where scattering maps along homoclinic channels to normally hyperbolic invariant manifolds can sometimes be varied by perturbing the Hamiltonian near the channel. To convert the present conditional transfer principle into an unconditional generic Hamiltonian theorem, one would need to prove sufficiently strong transversality, local-surjectivity, or openness properties for the induced map Φ . Establishing such properties is a substantial analytic problem that depends on the specific Hamiltonian family and lies beyond the purely topological methods of the present paper.

Further directions

Several natural extensions suggest themselves. First, the private-edge scheme of Definition 6.2 could potentially be relaxed further: instead of requiring a private edge that appears exactly once in one loop and not at all in the other, one might allow more general overlap patterns, provided the resulting dependence of the return maps on the distinguished edge parameters remains sufficiently non-degenerate.

Second, the present results are purely topological: they establish open-set transitivity and residual dense admissible branches, but do not address invariant measures, ergodicity, or mixing for any associated skew-product or random branch dynamics. It would be interesting to know whether the present Baire-category framework can be combined with measure-theoretic or probabilistic methods in order to obtain ergodic information in graph-directed settings.

Third, in concrete Hamiltonian families one may ask for quantitative versions of the topological coverage results proved here. For example, one could study rates at which admissible branches visit prescribed windows or atoms, and investigate whether such rates can be related to quantitative recurrence estimates or other effective dynamical bounds in specific models.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, numerical interpretations, and bibliographic information were subsequently reviewed and edited by the author, who takes full responsibility for the content of the manuscript.

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